

Hilbert action

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1 HILBERT ACTION

Hilbert action

$$S_H = \int \sqrt{-g} R d^n x \quad (1)$$

Take variation of metric $\delta g_{\mu\nu}$. Then the metric under variation is

$$g'_{\mu\nu} = (g_{\mu\nu} + \delta g_{\mu\nu}) \quad (2)$$

then the inverse is

$$g'^{\mu\nu} = (g^{\mu\nu} + \delta g^{\mu\nu}) \quad (3)$$

if we act it on the metric

$$\delta^\mu{}_\nu = g'^{\mu\rho} g'_{\rho\nu} = (g^{\mu\rho} + \delta g^{\mu\rho})(g_{\rho\nu} + \delta g_{\rho\nu}) \quad (4)$$

$$= g^{\mu\rho} g_{\rho\nu} + g_{\rho\nu} \delta g^{\mu\rho} + g^{\mu\rho} \delta g_{\rho\nu} + \mathcal{O}(\delta^2) \quad (5)$$

$$= \delta^\mu{}_\nu + g_{\rho\nu} \delta g^{\mu\rho} + g^{\mu\rho} \delta g_{\rho\nu} + \mathcal{O}(\delta^2) \quad (6)$$

then by neglecting $\mathcal{O}(\delta^2)$ term

$$0 = g_{\rho\nu} \delta g^{\mu\rho} + g^{\mu\rho} \delta g_{\rho\nu} = g_{\rho\nu} g^{\mu\lambda} g_{\lambda\sigma} \delta g^{\sigma\rho} + g^{\mu\rho} \delta g_{\rho\nu} \quad (7)$$

$$= g^{\mu\rho} (g_{\rho\sigma} g_{\nu\lambda} \delta g^{\sigma\lambda} + \delta g_{\rho\nu}) \quad (8)$$

Since this is true for any point and indeices of metric, by variation of inverse metric

$$\delta g_{\mu\nu} = -g_{\mu\sigma} g_{\nu\lambda} \delta g^{\sigma\lambda}. \quad (9)$$

The variation of Hilbert action is

$$\delta S_H = \int \delta \sqrt{-g} R d^n x + \int \sqrt{-g} \delta R d^n x \quad (10)$$

$$= \int \delta \sqrt{-g} R d^n x + \int \sqrt{-g} \delta (g^{\mu\nu} R_{\mu\nu}) d^n x \quad (11)$$

$$= \int \delta \sqrt{-g} R d^n x + \int \sqrt{-g} \delta g^{\mu\nu} R_{\mu\nu} d^n x + \int \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} d^n x \quad (12)$$

$$(13)$$

Say

$$\delta S_1 = \int \delta \sqrt{-g} R d^n x \quad (14)$$

$$\delta S_2 = \int \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} d^n x \quad (15)$$

$$\delta S_3 = \int \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} d^n x \quad (16)$$

For δS_1 ,

$$\delta \sqrt{-g} = -\frac{1}{2} \frac{\delta g}{\sqrt{-g}}. \quad (17)$$

We now use two lemmas and a proposition

Lemma 1.1. *If M is a diagonalizable matrix, and its eigen values are $\{\alpha_i\}$, the matrix P transform M to diagonal matrix, then*

$$\det(M) = \prod_i \alpha_i \quad (18)$$

Proof. since determinant is invariant

$$\det(P^{-1}MP) = \det(P^{-1}) \det(M) \det(P) = \det(M) \quad (19)$$

then, because $P^{-1}MP$ is diagonal,

$$\det(M) = \prod_i \alpha_i \quad (20)$$

Lemma 1.2. *If M is a diagonalizable matrix, then*

$$\det(\exp(M)) = \exp(\text{Tr}(M)) \quad (21)$$

Proof. By definition

$$\exp(M) = \sum_{k=0}^{\infty} \frac{M^k}{k!} \quad (22)$$

then eigenvalues of $\exp(M)$ are $\{\exp(\alpha_i)\}$. Thus by lemma 1.1

$$\det(\exp(M)) = \prod_i \exp(\alpha_i) = \exp\left(\sum_i \alpha_i\right) \quad (23)$$

Since trace is invariant of basis, that $\text{Tr}(M) = \text{Tr}(P^{-1}MP) = \sum_i \alpha_i$,

$$\det(\exp(M)) = \exp(\text{Tr}(M)) \quad (24)$$

Proposition 1.3.

$$\log(\det(M)) = \text{Tr}(\log(M)) \quad (25)$$

Proof. Define $\log(M)$ by $\exp(\log(M)) = M$. By lemma 1.2, and replacing $M \rightarrow \log(M)$

$$\det(M) = \exp(\text{Tr}(\log(M))). \quad (26)$$

By taking logarithm both side

$$\log(\det(M)) = \text{Tr}(\log(M)) \quad (27)$$

We can use this proposition for metric, because metric is generally diagonalizable for its symmetry;

$$\log(g) = \text{Tr}(\log(g_{\mu\nu} dx^\mu \otimes dx^\nu)). \quad (28)$$

If we write matrix of metric by g^M

$$\delta \log(g) = \frac{1}{g} \delta g \quad (29)$$

and

$$\delta \text{Tr}(\log(g^M)) = \text{Tr}(\log(g^M + \delta g^M)) - \text{Tr}(\log g^M) \quad (30)$$

$$= \text{Tr}(\log g^M + \log(E + (g^M)^{-1} \delta g^M)) - \text{Tr}(\log(g^M)) \quad (31)$$

$$= \text{Tr}(\log(E + (g^M)^{-1} \delta g^M)) \quad (32)$$

$$= \text{Tr}((g^M)^{-1} \delta g^M) + \mathcal{O}((\delta g^M)^2), \quad (33)$$

then

$$\frac{1}{g} \delta g = g^{\mu\nu} \delta g_{\mu\nu} \quad (34)$$

and

$$\delta g = g(g^{\mu\nu} \delta g_{\mu\nu}) = -g(g^{\mu\nu} g_{\mu\rho} g_{\nu\sigma} \delta g^{\rho\sigma}) = -g(g_{\mu\nu} \delta g^{\mu\nu}), \quad (35)$$

now go back to $\delta\sqrt{-g}$,

$$\delta\sqrt{-g} = \frac{g}{2\sqrt{-g}} g_{\mu\nu} \delta g^{\mu\nu} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} \quad (36)$$

and δS_1 is

$$\delta S_1 = \int dx^n \sqrt{-g} \left(-\frac{1}{2} R g_{\mu\nu} \delta g^{\mu\nu} \right) \quad (37)$$

For δS_3 ,

$$\delta R_{\mu\nu} = \delta(g^{\sigma\lambda} g_{\lambda\rho} R^\rho_{\mu\sigma\nu}) \quad (38)$$

$$= g_{\lambda\rho} R^\rho_{\mu\sigma\nu} \delta g^{\sigma\lambda} + g^{\sigma\lambda} R^\rho_{\mu\sigma\nu} \delta g_{\lambda\rho} + g^{\sigma\lambda} g_{\lambda\rho} \delta R^\rho_{\mu\sigma\nu} \quad (39)$$

$$= g_{\lambda\rho} R^\rho_{\mu\sigma\nu} \delta g^{\sigma\lambda} - g^{\sigma\lambda} R^\rho_{\mu\sigma\nu} g_{\lambda\alpha} g_{\rho\beta} \delta g^{\alpha\beta} + g^{\sigma\lambda} g_{\lambda\rho} \delta R^\rho_{\mu\sigma\nu} \quad (40)$$

$$= g_{\lambda\rho} R^\rho_{\mu\sigma\nu} \delta g^{\sigma\lambda} - g_{\rho\lambda} R^\rho_{\mu\sigma\nu} \delta g^{\sigma\lambda} + g^{\sigma\lambda} g_{\lambda\rho} \delta R^\rho_{\mu\sigma\nu} \quad (41)$$

$$= g^{\sigma\lambda} g_{\lambda\rho} \delta R^\rho_{\mu\sigma\nu} \quad (42)$$

$$(43)$$

(note that this is trivial calculation because $\delta(\delta^\mu_\nu) = 0$)

Since Riemann tensor is

$$R^\rho_{\mu\sigma\nu} = \partial_\sigma \Gamma^\rho_{\nu\mu} - \partial_\nu \Gamma^\rho_{\sigma\mu} + \Gamma^\rho_{\sigma\lambda} \Gamma^\lambda_{\nu\mu} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\sigma\mu}, \quad (44)$$

$$\delta R^\rho_{\mu\sigma\nu} = \partial_\sigma \delta \Gamma^\rho_{\nu\mu} - \partial_\nu \delta \Gamma^\rho_{\sigma\mu} + (\delta \Gamma^\rho_{\sigma\lambda}) \Gamma^\lambda_{\nu\mu} + \Gamma^\rho_{\sigma\lambda} (\delta \Gamma^\lambda_{\nu\mu}) - (\delta \Gamma^\rho_{\nu\lambda}) \Gamma^\lambda_{\sigma\mu} - \Gamma^\rho_{\nu\lambda} (\delta \Gamma^\lambda_{\sigma\mu}) \quad (45)$$

$$= \left(\partial_\sigma (\delta \Gamma^\rho_{\nu\mu}) + \Gamma^\rho_{\sigma\lambda} (\delta \Gamma^\lambda_{\nu\mu}) - \Gamma^\lambda_{\sigma\mu} (\delta \Gamma^\rho_{\nu\lambda}) - \Gamma^\lambda_{\sigma\nu} (\delta \Gamma^\rho_{\lambda\mu}) \right) + \Gamma^\lambda_{\sigma\nu} (\delta \Gamma^\rho_{\lambda\mu}) \\ - \left(\partial_\nu (\delta \Gamma^\rho_{\sigma\mu}) + \Gamma^\rho_{\nu\lambda} (\delta \Gamma^\lambda_{\sigma\mu}) - \Gamma^\lambda_{\nu\mu} (\delta \Gamma^\rho_{\sigma\lambda}) - \Gamma^\lambda_{\nu\sigma} (\delta \Gamma^\rho_{\lambda\mu}) \right) - \Gamma^\lambda_{\nu\sigma} (\delta \Gamma^\rho_{\lambda\mu}) \quad (46)$$

$$= \nabla_\sigma (\delta \Gamma^\rho_{\nu\mu}) - \nabla_\nu (\delta \Gamma^\rho_{\sigma\mu}) + \Gamma^\lambda_{\sigma\nu} (\delta \Gamma^\rho_{\lambda\mu}) - \Gamma^\lambda_{\nu\sigma} (\delta \Gamma^\rho_{\lambda\mu}) \quad (47)$$

By torsion free condition

$$\delta R^\rho_{\mu\sigma\nu} = \nabla_\sigma (\delta \Gamma^\rho_{\nu\mu}) - \nabla_\nu (\delta \Gamma^\rho_{\sigma\mu}). \quad (48)$$

Thus,

$$\delta R_{\mu\nu} = \nabla_\lambda (\delta \Gamma^\lambda_{\nu\mu}) - \nabla_\nu (\delta \Gamma^\lambda_{\lambda\mu}) \quad (49)$$

and

$$g^{\mu\nu} \delta R_{\mu\nu} = g^{\mu\nu} \nabla_\lambda (\delta \Gamma^\lambda_{\nu\mu}) - g^{\mu\nu} \nabla_\nu (\delta \Gamma^\lambda_{\lambda\mu}) \quad (50)$$

$$= \nabla_\lambda (g^{\mu\nu} \delta \Gamma^\lambda_{\nu\mu}) - \nabla_\nu (g^{\mu\nu} \delta \Gamma^\lambda_{\lambda\mu}) \quad (51)$$

$$= \nabla_\lambda (g^{\mu\nu} \delta \Gamma^\lambda_{\nu\mu} - g^{\mu\lambda} \delta \Gamma^\sigma_{\sigma\mu}) \quad (52)$$

Consider variation of Christoffel symbol is

$$\delta\Gamma^\rho_{\mu\nu} = \frac{1}{2}\delta(g^{\rho\lambda}(\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu})) \quad (53)$$

$$= \frac{1}{2}(\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu})\delta g^{\rho\lambda} + \frac{1}{2}g^{\rho\lambda}(\partial_\mu(\delta g_{\nu\lambda}) + \partial_\nu(\delta g_{\lambda\mu}) - \partial_\lambda(\delta g_{\mu\nu})) \quad (54)$$

$$= \frac{1}{2}(\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu})\delta g^{\rho\lambda} - \frac{1}{2}g^{\rho\lambda}(\partial_\mu(g_{\nu\alpha}g_{\lambda\beta}\delta g^{\alpha\beta}) + \partial_\nu(g_{\lambda\alpha}g_{\mu\beta}\delta g^{\alpha\beta}) - \partial_\lambda(g_{\mu\alpha}g_{\nu\beta}\delta g^{\alpha\beta})) \quad (55)$$

$$= \frac{1}{2}(\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu})\delta g^{\rho\lambda} - \frac{1}{2}g^{\rho\lambda}(\partial_\mu(g_{\nu\alpha}g_{\lambda\beta}) + \partial_\nu(g_{\lambda\alpha}g_{\mu\beta}) - \partial_\lambda(g_{\mu\alpha}g_{\nu\beta}))\delta g^{\alpha\beta} - \frac{1}{2}g^{\rho\lambda}(g_{\nu\alpha}g_{\lambda\beta}\partial_\mu(\delta g^{\alpha\beta}) + g_{\lambda\alpha}g_{\mu\beta}\partial_\nu(\delta g^{\alpha\beta}) - g_{\mu\alpha}g_{\nu\beta}\partial_\lambda(\delta g^{\alpha\beta})) \quad (56)$$

$$= \frac{1}{2}(\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu})\delta g^{\rho\lambda} - \frac{1}{2}g^{\rho\lambda}(g_{\nu\alpha}\partial_\mu(g_{\lambda\beta}) + g_{\mu\beta}\partial_\nu(g_{\lambda\alpha}) - g_{\mu\alpha}\partial_\lambda(g_{\nu\beta}))\delta g^{\alpha\beta} - \frac{1}{2}g^{\rho\lambda}(g_{\lambda\beta}\partial_\mu(g_{\nu\alpha}) + g_{\lambda\alpha}\partial_\nu(g_{\mu\beta}) - g_{\nu\beta}\partial_\lambda(g_{\mu\alpha}))\delta g^{\alpha\beta} - \frac{1}{2}(g_{\nu\alpha}\partial_\mu(\delta g^{\alpha\rho}) + g_{\mu\beta}\partial_\nu(\delta g^{\rho\beta}) - g_{\mu\alpha}g_{\nu\beta}g^{\rho\lambda}\partial_\lambda(\delta g^{\alpha\beta})) \quad (57)$$

$$= -\frac{1}{2}(\partial_\lambda g_{\mu\nu})\delta g^{\rho\lambda} - \frac{1}{2}g^{\rho\lambda}(g_{\nu\alpha}\partial_\mu(g_{\lambda\beta}) + g_{\mu\beta}\partial_\nu(g_{\lambda\alpha}) - g_{\mu\alpha}\partial_\lambda(g_{\nu\beta}))\delta g^{\alpha\beta} - \frac{1}{2}g^{\rho\lambda}(-g_{\nu\beta}\partial_\lambda(g_{\mu\alpha}))\delta g^{\alpha\beta} - \frac{1}{2}(g_{\nu\alpha}\partial_\mu(\delta g^{\alpha\rho}) + g_{\mu\beta}\partial_\nu(\delta g^{\rho\beta}) - g_{\mu\alpha}g_{\nu\beta}g^{\rho\lambda}\partial_\lambda(\delta g^{\alpha\beta})) = -\frac{1}{2}g_{\nu\alpha}(\partial_\mu(\delta g^{\alpha\rho}) + (\delta g^{\alpha\beta})g^{\rho\lambda}\partial_\mu g_{\lambda\beta}) - \frac{1}{2}g_{\mu\beta}(\partial_\nu(\delta g^{\rho\beta}) + (\delta g^{\alpha\beta})g^{\rho\lambda}\partial_\nu g_{\lambda\alpha}) + \frac{1}{2}g^{\rho\lambda}(g_{\mu\alpha}g_{\nu\beta}\partial_\lambda(\delta g^{\alpha\beta}) + (\delta g^{\alpha\beta})g_{\mu\alpha}\partial_\lambda g_{\nu\beta} + (\delta g^{\alpha\beta})g_{\nu\beta}\partial_\lambda g_{\mu\alpha}) - \frac{1}{2}(\partial_\lambda g_{\mu\nu})\delta g^{\rho\lambda} \quad (58)$$

Here we check some covariant derivative,

$$C_1 = g_{\nu\alpha} \nabla_\mu (\delta g^{\alpha\rho}) = g_{\nu\alpha} \partial_\mu (\delta g^{\alpha\rho}) + g_{\nu\alpha} \Gamma_{\mu\kappa}^\alpha \delta g^{\kappa\rho} + g_{\nu\alpha} \Gamma_{\mu\kappa}^\rho \delta g^{\alpha\kappa} \quad (59)$$

$$C_2 = g_{\mu\beta} \nabla_\nu (\delta g^{\rho\beta}) = g_{\mu\beta} \partial_\nu (\delta g^{\rho\beta}) + g_{\mu\beta} \Gamma_{\nu\kappa}^\rho \delta g^{\kappa\beta} + g_{\mu\beta} \Gamma_{\nu\kappa}^\beta \delta g^{\rho\kappa} \quad (60)$$

$$C_3 = g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} \nabla_\lambda (\delta g^{\alpha\beta}) = g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} \partial_\lambda (\delta g^{\alpha\beta}) + g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} \Gamma_{\lambda\kappa}^\alpha \delta g^{\kappa\beta} + g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} \Gamma_{\lambda\kappa}^\beta \delta g^{\alpha\kappa} \quad (61)$$

$$\begin{aligned} C_1 - g_{\nu\alpha} \partial_\mu (\delta g^{\alpha\rho}) &= \frac{1}{2} g_{\nu\alpha} g^{\rho\gamma} (\partial_\mu g_{\gamma\kappa} + \partial_\kappa g_{\mu\gamma} - \partial_\gamma g_{\mu\kappa}) \delta g^{\alpha\kappa} \\ &\quad + \frac{1}{2} g_{\nu\alpha} g^{\alpha\gamma} (\partial_\mu g_{\gamma\kappa} + \partial_\kappa g_{\mu\gamma} - \partial_\gamma g_{\mu\kappa}) \delta g^{\kappa\rho} \end{aligned} \quad (62)$$

$$\begin{aligned} &= \frac{1}{2} g_{\nu\alpha} g^{\rho\gamma} (\partial_\mu g_{\gamma\kappa} + \partial_\kappa g_{\mu\gamma} - \partial_\gamma g_{\mu\kappa}) \delta g^{\alpha\kappa} \\ &\quad + \frac{1}{2} (\partial_\mu g_{\nu\kappa} + \partial_\kappa g_{\mu\nu} - \partial_\nu g_{\mu\kappa}) \delta g^{\kappa\rho} \end{aligned} \quad (63)$$

$$\begin{aligned} C_2 - g_{\mu\beta} \partial_\nu (\delta g^{\rho\beta}) &= \frac{1}{2} g_{\mu\beta} g^{\rho\gamma} (\partial_\nu g_{\gamma\kappa} + \partial_\kappa g_{\nu\gamma} - \partial_\gamma g_{\nu\kappa}) \delta g^{\kappa\beta} \\ &\quad + \frac{1}{2} g_{\mu\beta} g^{\beta\gamma} (\partial_\nu g_{\gamma\kappa} + \partial_\kappa g_{\nu\gamma} - \partial_\gamma g_{\nu\kappa}) \delta g^{\rho\kappa} \end{aligned} \quad (64)$$

$$\begin{aligned} &= \frac{1}{2} g_{\mu\beta} g^{\rho\gamma} (\partial_\nu g_{\gamma\kappa} + \partial_\kappa g_{\nu\gamma} - \partial_\gamma g_{\nu\kappa}) \delta g^{\kappa\beta} \\ &\quad + \frac{1}{2} (\partial_\nu g_{\mu\kappa} + \partial_\kappa g_{\nu\mu} - \partial_\mu g_{\nu\kappa}) \delta g^{\rho\kappa} \end{aligned} \quad (65)$$

$$\begin{aligned} C_3 - g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} \partial_\lambda (\delta g^{\alpha\beta}) &= \frac{1}{2} g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} g^{\alpha\gamma} (\partial_\lambda g_{\gamma\kappa} + \partial_\kappa g_{\lambda\gamma} - \partial_\gamma g_{\lambda\kappa}) \delta g^{\kappa\beta} \\ &\quad + \frac{1}{2} g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} g^{\beta\gamma} (\partial_\lambda g_{\gamma\kappa} + \partial_\kappa g_{\lambda\gamma} - \partial_\gamma g_{\lambda\kappa}) \delta g^{\alpha\kappa} \end{aligned} \quad (66)$$

$$\begin{aligned} &= \frac{1}{2} g_{\nu\beta} g^{\rho\lambda} (\partial_\lambda g_{\mu\kappa} + \partial_\kappa g_{\lambda\mu} - \partial_\mu g_{\lambda\kappa}) \delta g^{\kappa\beta} \\ &\quad + \frac{1}{2} g_{\mu\alpha} g^{\rho\lambda} (\partial_\lambda g_{\nu\kappa} + \partial_\kappa g_{\lambda\nu} - \partial_\nu g_{\lambda\kappa}) \delta g^{\alpha\kappa} \end{aligned} \quad (67)$$

$$\begin{aligned} &= \frac{1}{2} g_{\nu\alpha} g^{\rho\gamma} (\partial_\gamma g_{\mu\kappa} + \partial_\kappa g_{\gamma\mu} - \partial_\mu g_{\gamma\kappa}) \delta g^{\kappa\alpha} \\ &\quad + \frac{1}{2} g_{\mu\beta} g^{\rho\gamma} (\partial_\gamma g_{\nu\kappa} + \partial_\kappa g_{\gamma\nu} - \partial_\nu g_{\gamma\kappa}) \delta g^{\beta\kappa} \end{aligned} \quad (68)$$

Thus

$$\begin{aligned}
C_1 + C_2 - C_3 &= g_{\nu\alpha} \partial_\mu (\delta g^{\alpha\rho}) + g_{\nu\alpha} g^{\rho\gamma} (\partial_\mu g_{\gamma\kappa} - \partial_\gamma g_{\mu\kappa}) \delta g^{\alpha\kappa} \\
&\quad + g_{\mu\beta} \partial_\nu (\delta g^{\rho\beta}) + g_{\mu\beta} g^{\rho\gamma} (\partial_\nu g_{\gamma\kappa} - \partial_\gamma g_{\nu\kappa}) \delta g^{\kappa\beta} \\
&\quad - g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} \partial_\lambda (\delta g^{\alpha\beta}) \\
&\quad + \frac{1}{2} ((\partial_\mu g_{\nu\kappa} + \partial_\kappa g_{\mu\nu} - \partial_\nu g_{\mu\kappa}) \delta g^{\kappa\rho} + (\partial_\nu g_{\mu\kappa} + \partial_\kappa g_{\nu\mu} - \partial_\mu g_{\nu\kappa}) \delta g^{\rho\kappa}) \quad (69)
\end{aligned}$$

$$\begin{aligned}
&= g_{\nu\alpha} \partial_\mu (\delta g^{\alpha\rho}) + g_{\nu\alpha} g^{\rho\lambda} (\partial_\mu g_{\lambda\kappa} - \partial_\lambda g_{\mu\kappa}) \delta g^{\alpha\kappa} \\
&\quad + g_{\mu\beta} \partial_\nu (\delta g^{\rho\beta}) + g_{\mu\beta} g^{\rho\lambda} (\partial_\nu g_{\lambda\kappa} - \partial_\lambda g_{\nu\kappa}) \delta g^{\kappa\beta} \\
&\quad - g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} \partial_\lambda (\delta g^{\alpha\beta}) \\
&\quad + (\partial_\kappa g_{\mu\nu}) \delta g^{\kappa\rho} \quad (70)
\end{aligned}$$

$$\begin{aligned}
&= g_{\nu\alpha} \partial_\mu (\delta g^{\alpha\rho}) + g_{\nu\alpha} g^{\rho\lambda} (\partial_\mu g_{\lambda\beta}) (\delta g^{\alpha\beta}) - g_{\nu\alpha} g^{\rho\lambda} (\partial_\lambda g_{\mu\beta}) \delta g^{\alpha\beta} \\
&\quad + g_{\mu\beta} \partial_\nu (\delta g^{\rho\beta}) + g_{\mu\beta} g^{\rho\lambda} (\partial_\nu g_{\lambda\alpha}) (\delta g^{\alpha\beta}) - g_{\mu\beta} g^{\rho\lambda} (\partial_\lambda g_{\nu\alpha}) \delta g^{\alpha\beta} \\
&\quad - g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} \partial_\lambda (\delta g^{\alpha\beta}) \\
&\quad + (\partial_\lambda g_{\mu\nu}) \delta g^{\lambda\rho} \quad (71)
\end{aligned}$$

$$\begin{aligned}
&= g_{\nu\alpha} \partial_\mu (\delta g^{\alpha\rho}) + g_{\nu\alpha} g^{\rho\lambda} (\partial_\mu g_{\lambda\beta}) \delta g^{\alpha\beta} \\
&\quad + g_{\mu\beta} \partial_\nu (\delta g^{\rho\beta}) + g_{\mu\beta} g^{\rho\lambda} (\partial_\nu g_{\lambda\alpha}) \delta g^{\alpha\beta} \\
&\quad - g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} \partial_\lambda (\delta g^{\alpha\beta}) - g_{\nu\alpha} g^{\rho\lambda} (\partial_\lambda g_{\mu\beta}) \delta g^{\alpha\beta} - g_{\mu\beta} g^{\rho\lambda} (\partial_\lambda g_{\nu\alpha}) \delta g^{\alpha\beta} \\
&\quad + (\partial_\lambda g_{\mu\nu}) \delta g^{\rho\lambda} \quad (72)
\end{aligned}$$

By this we can see that,

$$\delta \Gamma^\rho_{\mu\nu} = -\frac{1}{2} (C_1 + C_2 - C_3) = -\frac{1}{2} [g_{\nu\lambda} \nabla_\mu (\delta g^{\lambda\rho}) + g_{\mu\lambda} \nabla_\nu (\delta g^{\rho\lambda}) - g_{\mu\alpha} g_{\nu\beta} g^{\rho\lambda} \nabla_\lambda (\delta g^{\alpha\beta})] \quad (73)$$

note that

$$\delta \Gamma^\lambda_{\nu\mu} = -\frac{1}{2} [g_{\mu\kappa} \nabla_\nu (\delta g^{\kappa\lambda}) + g_{\nu\kappa} \nabla_\mu (\delta g^{\lambda\kappa}) - g_{\nu\alpha} g_{\mu\beta} g^{\lambda\kappa} \nabla_\kappa (\delta g^{\alpha\beta})], \quad (74)$$

$$\delta \Gamma^\sigma_{\sigma\mu} = -\frac{1}{2} [g_{\mu\kappa} \nabla_\sigma (\delta g^{\kappa\sigma}) + g_{\sigma\kappa} \nabla_\mu (\delta g^{\sigma\kappa}) - g_{\sigma\alpha} g_{\mu\beta} g^{\sigma\kappa} \nabla_\kappa (\delta g^{\alpha\beta})]. \quad (75)$$

Then

$$\begin{aligned}
g^{\mu\nu} \delta \Gamma^\lambda_{\nu\mu} - g^{\mu\lambda} \delta \Gamma^\sigma_{\sigma\mu} &= -\frac{1}{2} [g^{\mu\nu} \{g_{\mu\kappa} \nabla_\nu (\delta g^{\kappa\lambda}) + g_{\nu\kappa} \nabla_\mu (\delta g^{\lambda\kappa}) - g_{\nu\alpha} g_{\mu\beta} g^{\lambda\kappa} \nabla_\kappa (\delta g^{\alpha\beta})\} \\
&\quad - g^{\mu\lambda} \{g_{\mu\kappa} \nabla_\sigma (\delta g^{\kappa\sigma}) + g_{\sigma\kappa} \nabla_\mu (\delta g^{\sigma\kappa}) - g_{\sigma\alpha} g_{\mu\beta} g^{\sigma\kappa} \nabla_\kappa (\delta g^{\alpha\beta})\}] \quad (76)
\end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{2} [\nabla_\nu (\delta g^{\nu\lambda}) + \nabla_\mu (\delta g^{\lambda\mu}) - g_{\mu\beta} g^{\lambda\kappa} \nabla_\kappa (\delta g^{\mu\beta}) \\
&\quad - \nabla_\sigma (\delta g^{\lambda\sigma}) - g^{\mu\lambda} g_{\sigma\kappa} \nabla_\mu (\delta g^{\sigma\kappa}) + \nabla_\kappa (\delta g^{\kappa\lambda})] \quad (77)
\end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{2} [\nabla_\nu (\delta g^{\nu\lambda}) + \nabla_\mu (\delta g^{\lambda\mu}) - g_{\mu\beta} g^{\lambda\kappa} \nabla_\kappa (\delta g^{\mu\beta}) \\
&\quad - \nabla_\mu (\delta g^{\lambda\mu}) - g^{\kappa\lambda} g_{\mu\beta} \nabla_\kappa (\delta g^{\mu\beta}) + \nabla_\kappa (\delta g^{\kappa\lambda})] \quad (78)
\end{aligned}$$

$$\begin{aligned}
&= g_{\mu\beta} g^{\lambda\kappa} \nabla_\kappa (\delta g^{\mu\beta}) - \nabla_\nu (\delta g^{\nu\lambda}) \quad (79)
\end{aligned}$$

and the variation with changing some dummy indices

$$g^{\mu\nu}\delta R_{\mu\nu} = \nabla_\lambda (g_{\mu\nu}g^{\lambda\sigma}\nabla_\sigma(\delta g^{\mu\nu}) - \nabla_\rho(\delta g^{\rho\lambda})) . \quad (80)$$

Hence we have

$$\delta S_3 = \int d^n x \sqrt{-g} \nabla_\lambda (g_{\mu\nu}g^{\lambda\sigma}\nabla_\sigma(\delta g^{\mu\nu}) - \nabla_\rho(\delta g^{\rho\lambda})) \quad (81)$$

and then

$$\delta S_H = \int d^n x \sqrt{-g} \left(-\frac{1}{2} R g_{\mu\nu} + R_{\mu\nu} \right) \delta g^{\mu\nu} + \int d^n x \sqrt{-g} \nabla_\lambda \{ g_{\mu\nu} g^{\lambda\sigma} \nabla_\sigma(\delta g^{\mu\nu}) - \nabla_\rho(\delta g^{\rho\lambda}) \} \quad (82)$$

By Stokes's theorem

$$\int_\Sigma d^n x \nabla_\mu V^\mu \sqrt{|g|} = \int_{\partial\Sigma} n_\mu V^\mu \sqrt{|\gamma|} d^{n-1}x, \quad (83)$$

the second integral in δS_H is

$$\int d^n x \sqrt{-g} \nabla_\lambda \{ g_{\mu\nu} g^{\lambda\sigma} \nabla_\sigma(\delta g^{\mu\nu}) - \nabla_\rho(\delta g^{\rho\lambda}) \} = (\text{Surface term}) = 0 \quad (84)$$

Finialy

$$\delta S_H = \int d^n x \sqrt{-g} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) \delta g^{\mu\nu} \quad (85)$$

and the equation we can obtain by functional derivarive is

$$\frac{1}{\sqrt{-g}} \frac{\delta S_H}{\delta g^{\mu\nu}} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \quad (86)$$

If it is vacuum the action is

$$S = S_H \quad (87)$$

and then we get Einstin's equation in vacuum;

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0 \quad (88)$$

2 IN PRESENCE OF MATTER

In presence of matters action is

$$S = \frac{1}{16\pi G} S_H + S_M \quad (89)$$

Then the equation by variation of metric is

$$\frac{1}{\sqrt{-g}} \frac{\delta S}{\delta g^{\mu\nu}} = \frac{1}{16\pi G} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) + \frac{1}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}} = 0 \quad (90)$$

The Einstein's equation is

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi G \left(-\frac{2}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}} \right) \quad (91)$$

define energy-momentum tensor by

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}} \quad (92)$$

Now complete Einstein's equation is

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu} \quad (93)$$

For example, if we adopt action for scalar field for S_M

$$S_M = \int dx^n \sqrt{-g} \left[-\frac{1}{2}g^{\mu\nu}(\nabla_\mu\phi)(\nabla_\nu\phi) - V(\phi) \right]. \quad (94)$$

Variation with respect to metric is

$$\delta S_M = \int dx^n \left[(\delta\sqrt{-g}) \left(-\frac{1}{2}(\nabla\phi)^2 - V(\phi) \right) - \frac{\sqrt{-g}}{2} \delta g^{\mu\nu} (\nabla_\mu\phi)(\nabla_\nu\phi) \right] \quad (95)$$

$$= \int dx^n \left[-\frac{\sqrt{-g}}{2} g_{\mu\nu} \left(-\frac{1}{2}(\nabla\phi)^2 - V(\phi) \right) - \frac{\sqrt{-g}}{2} (\nabla_\mu\phi)(\nabla_\nu\phi) \right] \delta g^{\mu\nu} \quad (96)$$

taking functional derivative

$$\frac{\delta S_M}{\delta g^{\mu\nu}} = -\frac{1}{2}\sqrt{-g} \left(g_{\mu\nu} \left(-\frac{1}{2}(\nabla\phi)^2 - V(\phi) \right) + (\nabla_\mu\phi)(\nabla_\nu\phi) \right) \quad (97)$$

substitute into definition of energy-momentum tensor,

$$T_{\mu\nu} = (\nabla_\mu\phi)(\nabla_\nu\phi) - \frac{1}{2}g_{\mu\nu}(\nabla\phi)^2 - g_{\mu\nu}V(\phi) \quad (98)$$

is energy-momentum tensor for scalar field.